Revisiting *Remarks*: the learning of English nominals

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1 Introduction. Generative linguistics has long been concerned with the descriptive adequacy and explanatory adequacy of theories. The typical approach is to propose a theory that first achieves descriptive adequacy, then discuss how it may relate to explanatory adequacy. The present paper illustrates an alternative approach: I propose an analysis that achieves explanatory adequacy, by showing how it can be learned by a child under a concrete theory of learning, then show that it also achieves descriptive adequacy.

I do so by considering the case of English nominalizations. I take as my starting point the observations in Chomsky 1970 on the distinction between gerundive nominals, as in (1b), and derived nominals, as in (1c).

(1) a. John refused the offer b. John's refusing the offer c. John's refusal of the offer

Gerundive and derived nominals differ in their productivity, internal syntax, and interpretation. These properties have motivated a range of subsequent work (e.g. Abney 1987, Grimshaw 1990, Marantz 1997, Alexiadou 2001, Wood 2023). Here, I approach the contrast by asking what a child learns about each construction from their input. I show how independently learnable rules combine to generate these nominals, and how their various properties follow from the results of learning.

2 The theory of learning. I adopt the view of learning underlying the Tolerance Principle (TP; Yang 2016). By default, a child memorizes the examples they encounter in their input as unproductive pieces of knowledge. The child may generalize these examples to a productive rule (i.e., a rule that applies to an open domain, rather than a list of memorized items) iff they receive sufficient evidence for the rule. The Tolerance Principle gives a concrete criterion for what constitutes sufficient evidence: (2). This view of the learning of a single rule extends naturally to the learning of a grammar, as the grammar can be understood as the set of learned rules.

(2) Let R be a rule defined over some domain of size N. Let M be the number of examples attested in the input that follow R. Then, the learner generalizes rule R over its domain if and only if $M \geq \gamma_N = N - \frac{N}{\ln N}$.

Since children learn the core grammar at a young age, we only need to check the productivity of rules on a small vocabulary. I thus use the most frequent 100 verbs in my corpus as the child's vocabulary of verbs. All counts and examples come from a 6.8-million-token corpus of child-directed speech (MacWhinney 2000).

3 Learning gerundive nominals. While the linguist takes "gerundive nominals" as the starting point of inquiry, the child has a different starting point: they are concerned with making sense of their input. Their input includes many phrases such as those in (3).

(3) a. you're [going to hurt me]
b. he came towards me [waving]
c. I enjoyed [reading your book]
d. thank you for [being our friend]
e. you were thinking about [being a teacher]
f. tell them about your [breaking your finger]

From these examples, the child can identify a unit that consistently recurs across varying environments. Internally, this unit involves *-ing*, along with material associated with a vP (e.g. direct objects). Externally, this unit is reused in different environments, with a distribution distinct from vP. This motivates a child to posit an abstract grammatical object: ingP.

The child can learn that ingP is productive over verbs. Of the most frequent 100 verbs, $M = 99$ of them are attested with *-ing*, easily surpassing the TP threshold for productivity, which is $\gamma_{100} = 79$. Furthermore, children can also learn that ingP contains vP structure: of the 83 transitive verbs within the top 100, $M = 80$ are attested with *-ing* and a direct object, surpassing the threshold of $\gamma_{83} = 65$. The child thus can learn the rule in (4).

(4) Rule: [$_{\text{ingP}}$ -ing vP] (productive over vP)

Once the child learns ingP, they can learn further rules that make reference to ingP. One such rule is that possessive determiners select ingP; this can be stated as in (5). This rule is productive over D_{poss}; of the 7 possessive determiners (*'s, my, your*, etc.), all $M = 7$ are attested with an ingP in the input.

(5) Rule: [$_{\text{DP}}$ D_{poss} ingP] (productive over D_{poss})

Together, the two rules in (4) and (5), along with the child's prior knowledge of basic categories, are sufficient to generate gerundive nominals. These components produce the structure of gerundive nominals as in (6). This analysis converges on standard analyses in which vP is contained inside a gerundive nominal (e.g. Abney 1987, Kratzer 1996, Marantz 1997), allowing us to understand their verbal properties.

(6) [$_{\text{DP}}$ John 's [$_{\text{ingP}}$ -ing [$_{\text{vP}}$ play the violin]]]

4 Learning derived nominals. For derived nominals, in contrast, there is no evidence to motivate a productive rule over vP. First, unlike *-ing*, other nominalizing suffixes are not productive. The most frequent nominalizing suffixes, *-tion*/*-ment*/*-al*, each only occur with one verb in the most frequent 100, *fixation*/*movement*/*trial*. This is far below any threshold for productivity. Second, the child has no evidence to posit that the phrases headed by these derived nouns contain vP structure; for example, derived nouns never take direct objects.

Thus, we can understand why derived nominals are not formed from vP: such a rule is simply not supported by the input. Instead, what the child does learn is the individual derived nouns as lexical items. They learn the category of words like *refusal* as nouns and the meaning of these words as predicates of events. Since derived nouns are of category N, they project an NP, and thus derived nominals have the internal syntax of an NP. Since derived nouns are event predicates, modifiers are interpreted as modifiers to the event, rather than an individual. Thus, these two pieces of lexical knowledge are sufficient to generate the basic syntax and semantics of derived nominals.

5 Discussion. I have thus provided an account of *how* gerundive and derived nominals are learned. This learning story resulted in an analysis of *what* exactly is (and is not) learned. This allows us to naturally understand the differences between gerundive and derived nominals, as they differ in what can be learned: the input supports rules that generate gerundive nominals productively from vP, while the input does not support such rules for derived nominals; instead, derived nominals result from learning individual derived nouns as lexical items.

More broadly, my account illustrates a general alternative approach to linguistic analysis: first propose an analysis that achieves explanatory adequacy, by showing how it can be learned by a child; then show that it achieves descriptive adequacy, by showing that the rules resulting from learning make the correct empirical predictions.

References. Abney (1987), *The English noun phrase in its sentential aspects*. Alexiadou (2001), *Functional structure in nominals*. Chomsky (1970), "Remarks on nominalization". Grimshaw (1990), *Argument structure*. Kratzer (1996), "Severing the external argument from its verb". MacWhinney (2000), *The CHILDES project*. Marantz (1997), "No escape from syntax", *Penn Working Papers in Linguistics*. Wood (2023), *Icelandic nominalizations and allosemy*. Yang (2016), *The price of linguistic productivity*.